PAUL BORGEN

AI Systems Engineer | *AWS Solutions Architect* | *Backend & Platorm Engineer*

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PROFESSIONAL SUMMARY

AI Systems Engineer with deep hands-on AWS expertise specializing in designing and building the production systems, infrastructure, and integration patterns that surround LLMs and generative AI. Proven ability to turn AI capabilities into enterprise-scale, secure, and operational platforms by connecting enterprise data sources, APIs, identity systems, and downstream workflows. Strong background in backend engineering, cloud-native architecture, and distributed systems. Experienced leading the design of HIPAA-compliant multi-tenant AI platforms and high-volume data pipelines, with a focus on reliability, scalability, security, and reusable architecture that can be federated across business units.

CORE SKILLS

- **AWS Plat orm & Architecture:** Amazon Bedrock, ECS/Fargate, Step Functions, OpenSearch Serverless, S3, Redis, IAM/RBAC, Lambda, Event-Driven Architectures, Terraform, CI/CD
- **AI / LLM Systems Engineering:** LangChain, LangGraph Agents, RAG & Vector Databases, Production LLM Integration, Agentic Workflows, Ollama
- **Backend & Distributed Systems:** Python, Golang, TypeScript, Java, APIs, Microservices, Data Pipelines, Enterprise Data Integration, Secure Workflows
- **Enterprise & Operational:** HIPAA Compliance, Authentication & Authorization, Observability, Infrastructure as Code, Scalable Production Systems, Mentoring
- **Additional:** Smart Contracts, Computer Vision, Automated Testing, Team Leadership

PROFESSIONAL EXPERIENCE

Alopex | Principal AI Engineer *05/2024 – Present*

- Architected and built a scalable, HIPAA-compliant, multi-tenant AI platform on AWS that operationalizes LLM-based solutions for clinical care coordination across medical organizations.
- Designed and implemented an AI-powered clinical chat interface using Amazon Bedrock, LangChain, and LangGraph agents running on AWS Fargate (ECS). Integrated OpenSearch Serverless for embeddings, S3 for document processing, Redis caching, and Step Functions for orchestration—delivering real-time, context-rich patient summaries that improve decision-making speed.
- Built the surrounding production systems and integration patterns (secure authentication, role-based access control, intelligent workflows for referrals, eligibility verification, and care team coordination) so LLM capabilities could reliably consume enterprise data and return AI-generated outputs into downstream systems.
- Focused on enterprise-scale architecture: reliability, security, maintainability, and operational readiness of the full system around the AI models rather than model development itself.
- Partnered with cross-functional teams to translate clinical use cases into reusable, scalable AI platform capabilities.

Mintra | Co-Founder / CTO *07/2021 – 05/2024*

- Led architecture and engineering of a high-scale NFT marketplace platform, including design and implementation of an LLM-powered recommendation engine that integrated user behavior, collection traits, and on-chain activity to deliver personalized suggestions.
- Architected data pipelines and distributed locking workflows that processed millions of NFTs and tens of thousands of collections in a deterministic, scalable manner—demonstrating ability to connect and transform complex data sources into production AI systems.
- Deployed and operated Ollama-based models on RunPod for the recommendation system while ensuring full smart-contract audits (Foundry/Hardhat) and production security.

- Leveraged modern AI development tools (Cursor, Claude) to accelerate engineering velocity and raise output quality across a global distributed team.

Precision Point Systems | Software Engineering Consultant *02/2020 – 07/2021*

- Evaluated and refined a sophisticated computer vision system for real-time football tracking using C/C++/GoLang, high-resolution cameras, and custom AI models with OpenCV.
- Stabilized the system for live production use at UConn football games, achieving pixel-level (≈0.5 inch) measurement accuracy.

Honeywell | Staff / Senior Software Engineer *07/2012 – 02/2020*

- Designed and built backend systems, external interfaces, and data processing within metering infrastructures, focusing on performance, security, and scalability.
- Strengthened enterprise security (ESAPI) to reduce XSS/XSRF vulnerabilities, led security reviews from penetration testing, and implemented unit testing and CI pipelines (Jenkins).
- Migrated systems to Spring/Hibernate, optimized SQL and threading, migrated SVN to Git, and led distributed teams in a Scrum environment.

EDUCATION & CERTIFICATIONS

Central Connecticut State University — Bachelor of Science, Computer Science

- **AWS Certified Generative AI Developer – Professional**
- AWS Certified Solutions Architect – Associate
- AWS Certified Cloud Practitioner
- Certified Ethical Hacker, EC-Council

PROFILES

GitHub: github.com/pborgen | Portfolio: paulborgen.com